

Software Requirement Specification

By Plummetarians : Tales of the นักเก็บกวาด(s)

Chosen Irl Challenge: Project D: Course Support & Activity Dashboard



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Table of Contents

1. Target Users.....	1
2. The System Goal.....	1
3. Goal Refinement via KAOS:.....	2
4. Use Case Diagram.....	3
5. User Stories.....	4
6. User Requirement Specification.....	6
7. Activity Diagram.....	7
AD-1: Lecturer and TAs account creation and login.....	7
AD-2: Course creation and management.....	8
AD-3: Lecturer Publishes Announcements.....	9
AD-4: Course's tasks creation.....	10
AD-5: TAs task handling.....	11
AD-6: Course's tasks management.....	12
AD-7: Consultant Appointment and Logging.....	13
AD-8: FAQ Log.....	14
8. System Requirement Specification.....	15
9. Software Architecture.....	16
10. Data Storage Technology.....	18
11. Development Environment.....	19
12. Sequence Diagrams for All SRS.....	20
SQD-1: Course Creation and Management.....	20
SQD-2: Announcement Creation and Management.....	21
SQD-3: Course tasks management.....	21
SQD-4: FAQs creation.....	22
SQD-5: Consultation appointment and Personalized Calendar.....	22
SQD-6: Lecturer/TAs Overview dashboard.....	23
SQD-7: Google Authentication to Dashboard.....	23
13. Traceability Matrix.....	24

1. Target Users

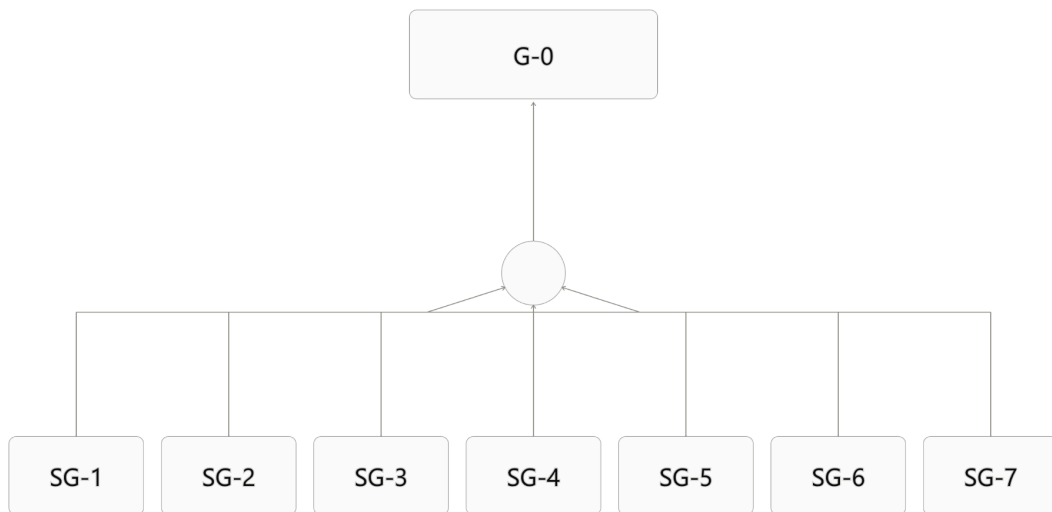
- **Lecturer (Admin):** Create and oversee the course progress, publish an announcement and assign TAs with a task and follow up with them.
- **Teacher Assistants (TAs):** Track and update their own assignment, lab session, and consultation appointment. Manages and updates FAQ for the course.
- **Students:** View FAQs that are updated by the Lecturers and TAs. Create an appointment for a consultation session with TA/Lecturer.

2. The System Goal

G-0: To centralize communication, task tracking, and progression report between lecturers and TA while providing consultant appointment and FAQs answer to students.

Currently, the lecturers and TAs are using fragmented communication and task tracking applications, resulting in the TAs struggling with the flow of the courses and the Lecturer having no way to view progress of each task other than directly asking the TAs for updates on the task. Additionally, TAs are also burdened with student consultation and questions that can sometimes be similar to each other, but due to the lack of centralized FAQs resources, they have to answer them every time. This project aims to centralize the communication, tracking and reflecting the progress of tasks for Lecturers and TAs to be on the same page. Moreover, providing students with FAQs answers and an appointment system for students to use.

3. Goal Refinement via KAOS:



- **Sub-Goal 1 (SG-1)**

Description: Achieve: Provide Lecturers with effective course and user management.

Rationale: Lecturers need to be able to create/manage courses and also manage accessibility to other Lecturers and TAs

- **Sub-Goal 2 (SG-2)**

Description: Achieve: Provide timely course announcements to TAs.

Rationale: TAs need to receive important course announcements from the lecturer via the dashboard and email

- **Sub-Goal 3 (SG-3)**

Description: Achieve: Provide effective task and schedule tracking for TAs.

Rationale: Lecturers need to create, assign, schedule, and monitor tasks, including managing their importance and risk of incompleteness.

- **Sub-Goal 4 (SG-4)**

Description: Maintain well-structured FAQs and information.

Rationale: FAQ content and activity information need to remain well organized and accessible for TAs to support students and review relevant activity information.

- **Sub-Goal 5 (SG-5)**

Description: Manage consultation appointments and maintain consultation records.

Rationale: Lecturers and TAs need to efficiently manage consultation appointments and keep organized consultation records to support student inquiries and track consultation activities.

- **Sub-Goal 6 (SG-6)**

Description: Provide Lecturers and TAs with a personalized overview of their assigned work and activities.

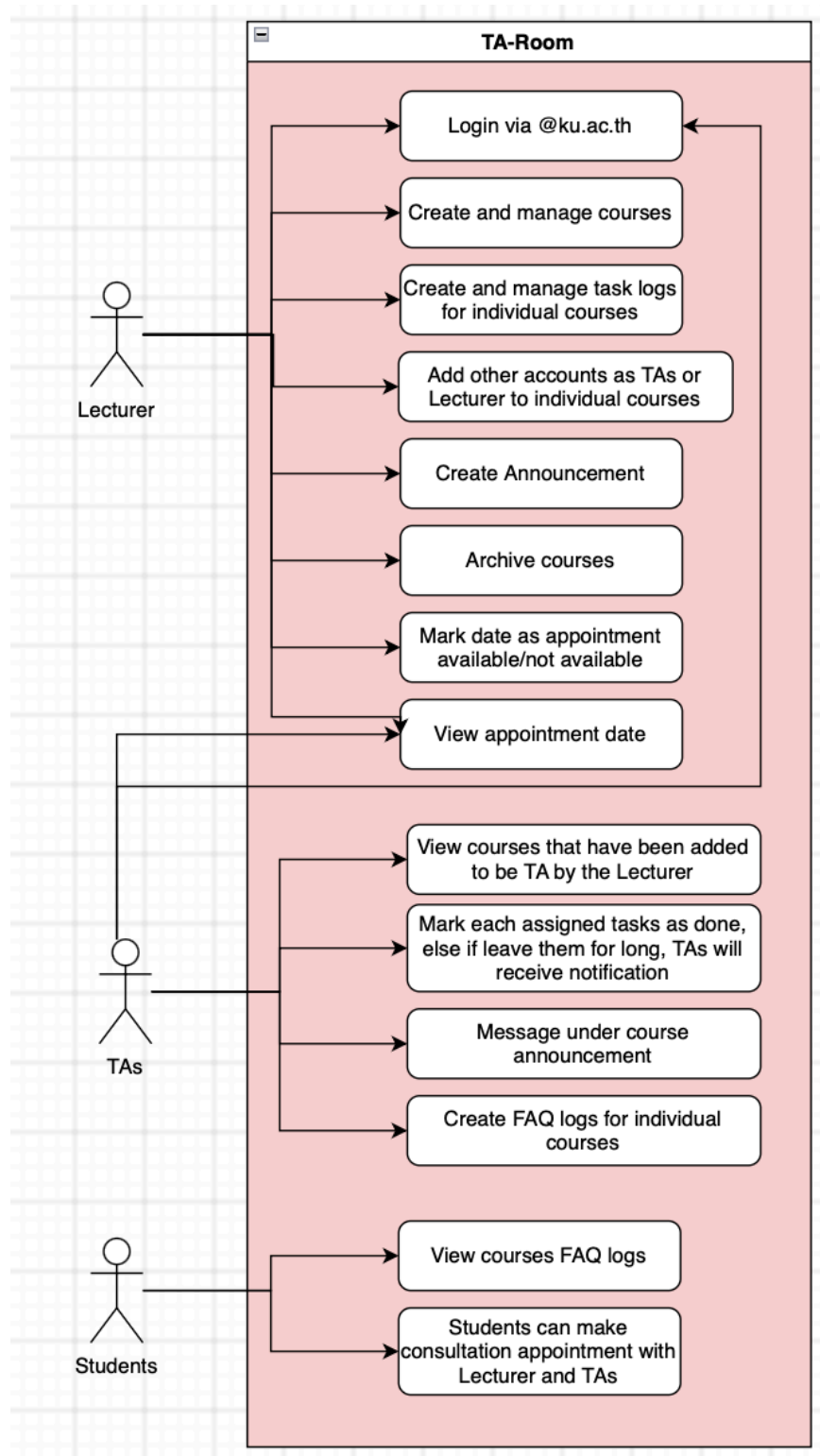
Rationale: Users need to view, track, and manage their own assignments and activities.

- **Sub-Goal 7 (SG-7)**

Description: Provide authentication via google account for dashboard access.

Rationale: Users need to be able to access the dashboard in an easy manner, google oauth2 can verify that users are studying/working in KU with only few interaction needed by the user.

4. Use Case Diagram



5. User Stories

User Story ID	Description	Acceptance Criteria
US-1	As a Lecturer, I want to create and manage a course with Lecturer/TA accounts, so that I can control who has access to teach or assist.	Creating a course requires a course name. Adding a TA/Lecturer requires a valid account. Removed accounts lose access immediately.
US-2	As a Lecturer, I want to publish an announcement to a course, so that TAs are notified without me messaging each of them individually.	Announcement requires title + body before publishing. Published announcements appear on TA dashboards. TAs with email set also receive an email.
US-3	As a Lecturer, I want to create, manage, and assign tasks with due dates to TAs, so that I can track progress without asking TAs directly for updates.	Task requires a title, description, due date, and optional assignee. A task is automatically flagged as "at risk" if (a) its due date has passed and status is not "Completed", or (b) it has had no status/progress update within N days. The flag is visible to the Lecturer on the task list and dashboard.
US-4	As a TAs, I want to update the progress of my assigned tasks, so that the Lecturer can see my work without a status meeting.	Task status can be changed by the assigned TA only. Consultation log entries require a course and date. Logs are visible only internally (not to students).
US-5	As a student, I want to view FAQs published by TAs/Lecturers, so that I can get answers without waiting for a reply.	published by TAs/Lecturers, so that I can get answers without waiting for a reply. FAQ list is publicly viewable per course. Only published FAQs (not drafts) are shown. Each FAQ shows question + answer text.
US-6	As a Lecturer and TAs, We want to be able to view the currently assigned tasks in a calendar view for a quick look to see what tasks need to be done before the deadline.	Create a calendar that tracks the tasks/schedules within that course with due dates and the description.
US-7	As a student, I want to conveniently consult with lecturers and TAs.	Students are able to make a consultation to Lecturers and TAs, selecting date and time for the appointment.

US-8	As TA and Lecturer, I want to be able to view the consultation made by students and get notified when the consultations are made.	In the course the TAs and Lecturers can view the consultation history that the students made and can see the details of the consultation, also they will get notified when the new consultation is made.
US-9	As a TA and Lecturer, I want to be able to log all the consultations that I already reviewed and create the FAQs pages for students to view.	After consultation, TAs and Lecturers can log the consultation to use it to create FAQs in the FAQ page to make students able to view later on.
US-10	As a TAs, I want to be able to search for specific tasks on each course.	TAs and Lecturers need to be able to search up for the task by letters.
US-11	As a Lecturer/TAs, We want to be able to login into the system using our google account.	TAs and Lecturers need to be able to login into the system using google oauth2 via @ ku.th email.

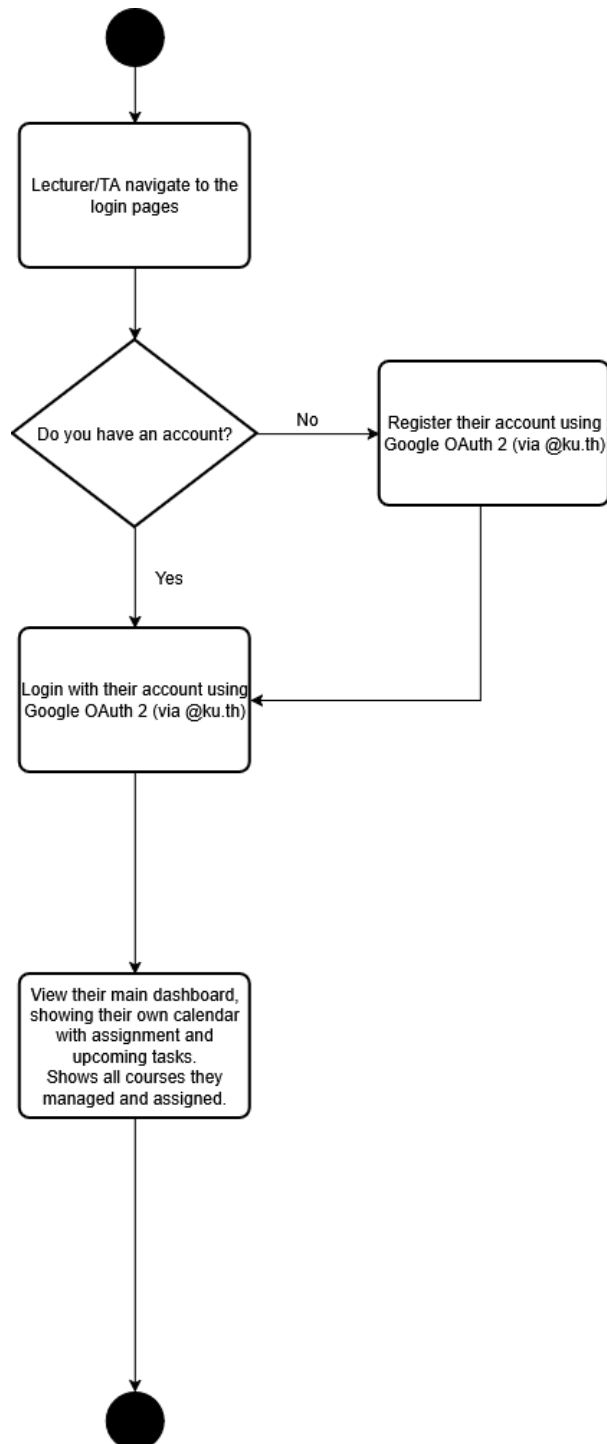
6. User Requirement Specification

ID	User Requirement
URS-1	Course Management : Lecturers can create and manage multiple courses. Creating/Updating access accounts of each additional Lecturer and TAs.
URS-2	Announcements : Lecturers can make an announcement for each course. Notifying TAs via their main dashboard and email. Both Lecturers and TAs can comment on announcements to discuss details.
URS-3	Task and Schedule : Lecturers can create, and manage tasks and schedule for TAs with optional specific user assignments. TAs able to update the progress of the assignment as well as searching for specific tasks.
URS-4	Consultant Appointment & Log : Students can create appointments for consultation. Lecturers and TAs can see the appointment dates and be notified for upcoming consultations. Additionally, lecturers and TAs can add previous consultations to the consultation log for future reviews
URS-5	FAQ Board : Students can view the FAQ board, lecturers and TAs can create/add frequently asked question notes for students to view.
URS-6	Personalized Overview : Lecturers and TAs have a main overview page to display tasks assigned to them over all courses. Additionally, there is an overview page for each course, showing only tasks assigned from that specific course.
URS-7	Google Authentication : Lecturers and TAs must be able to access the dashboard via google account OAuth 2 as requested by the stakeholder.

7. Activity Diagram

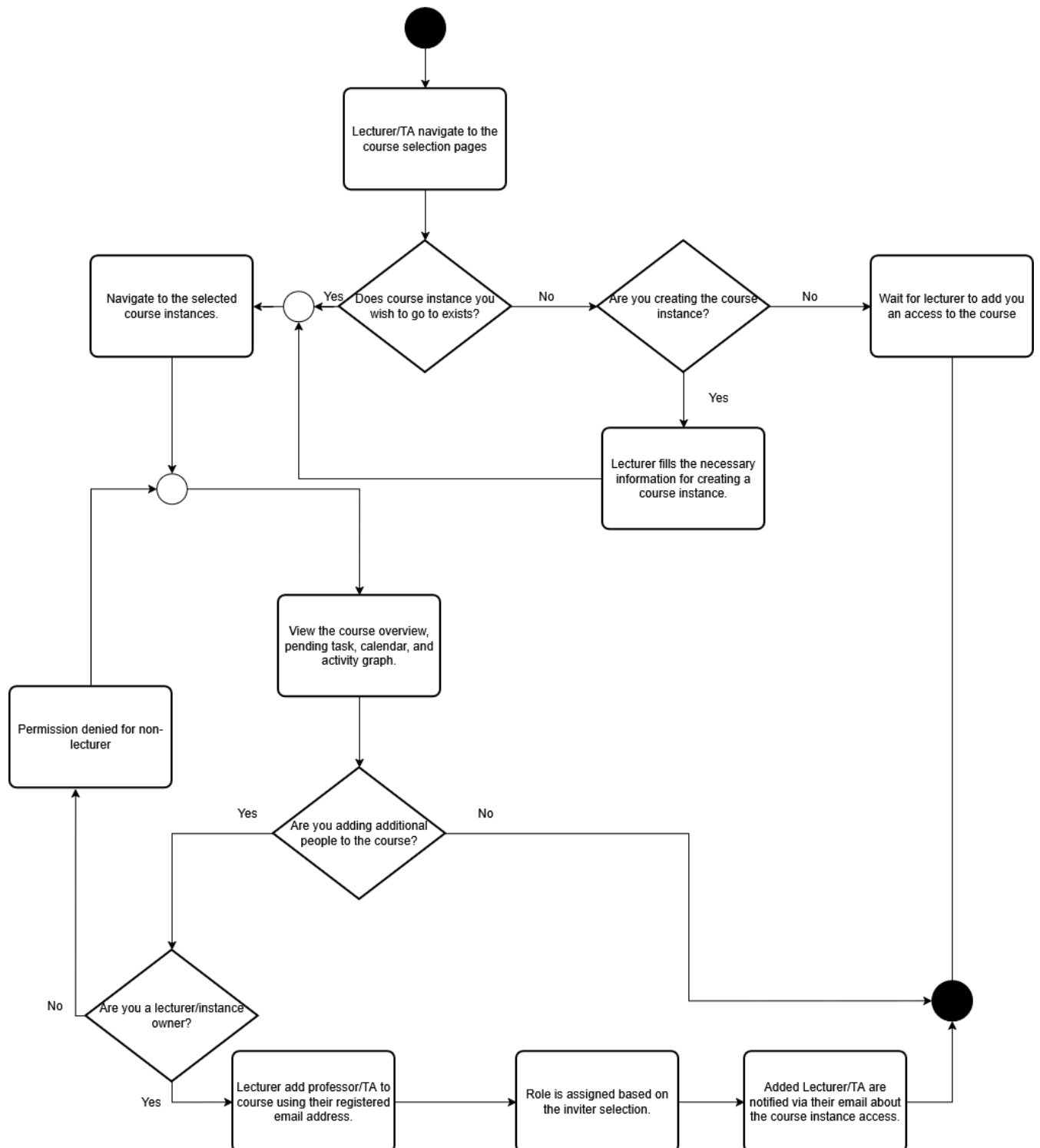
AD-1: Lecturer and TAs account creation and login.

This diagram shows the process for account registration and sign in process for Lecturers and TAs, which is converted to SRS-11



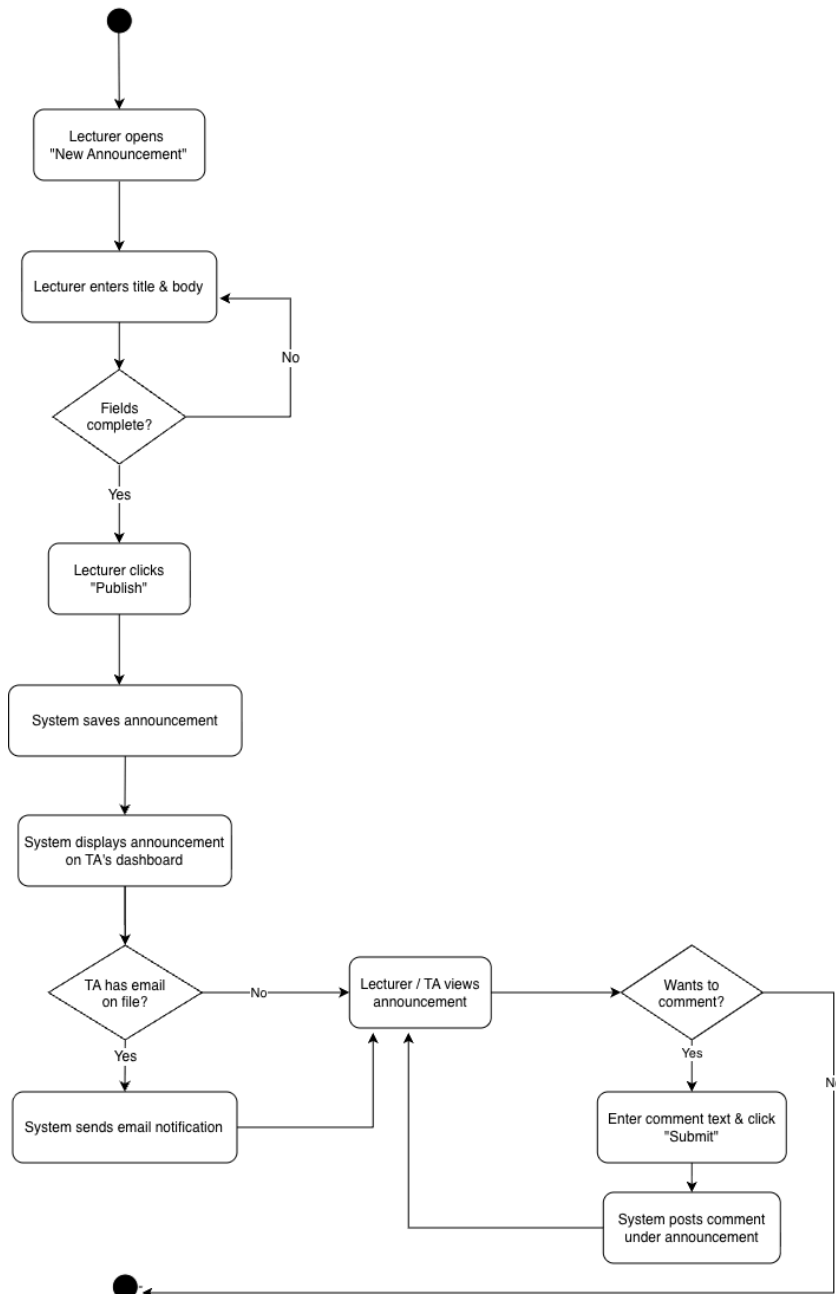
AD-2: Course creation and management.

This diagram shows the process for course instance creation and management, including user access management. Converted to SRS-1 SRS-2.



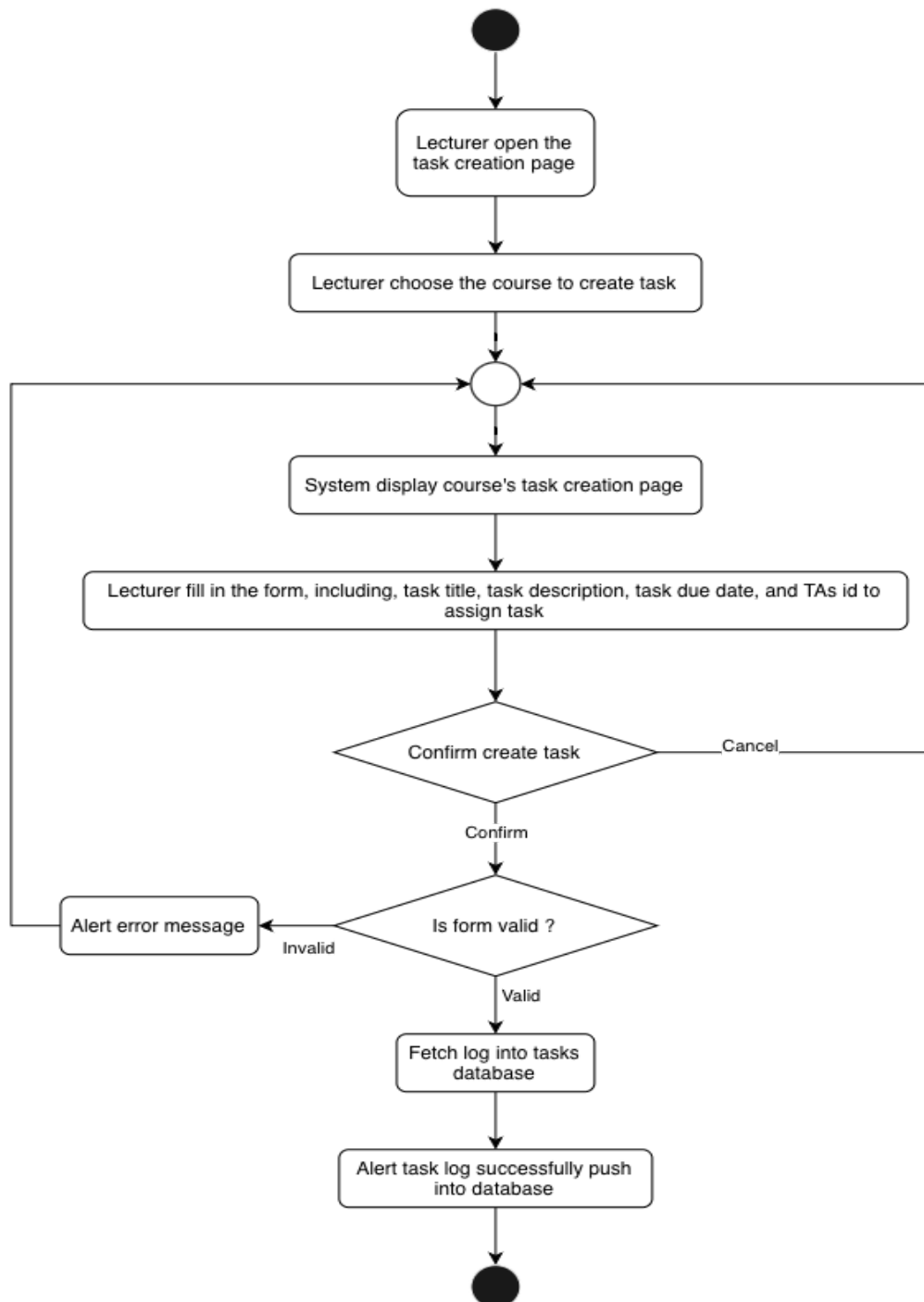
AD-3: Lecturer Publishes Announcements

This diagram traces the Lecturer's announcement- publishing flow from entering the title/body, through validation, to the system saving the announcement, pushing it to the TA dashboard, and conditionally emailing TAs who have an email on file. It was used to derive SRS-3.



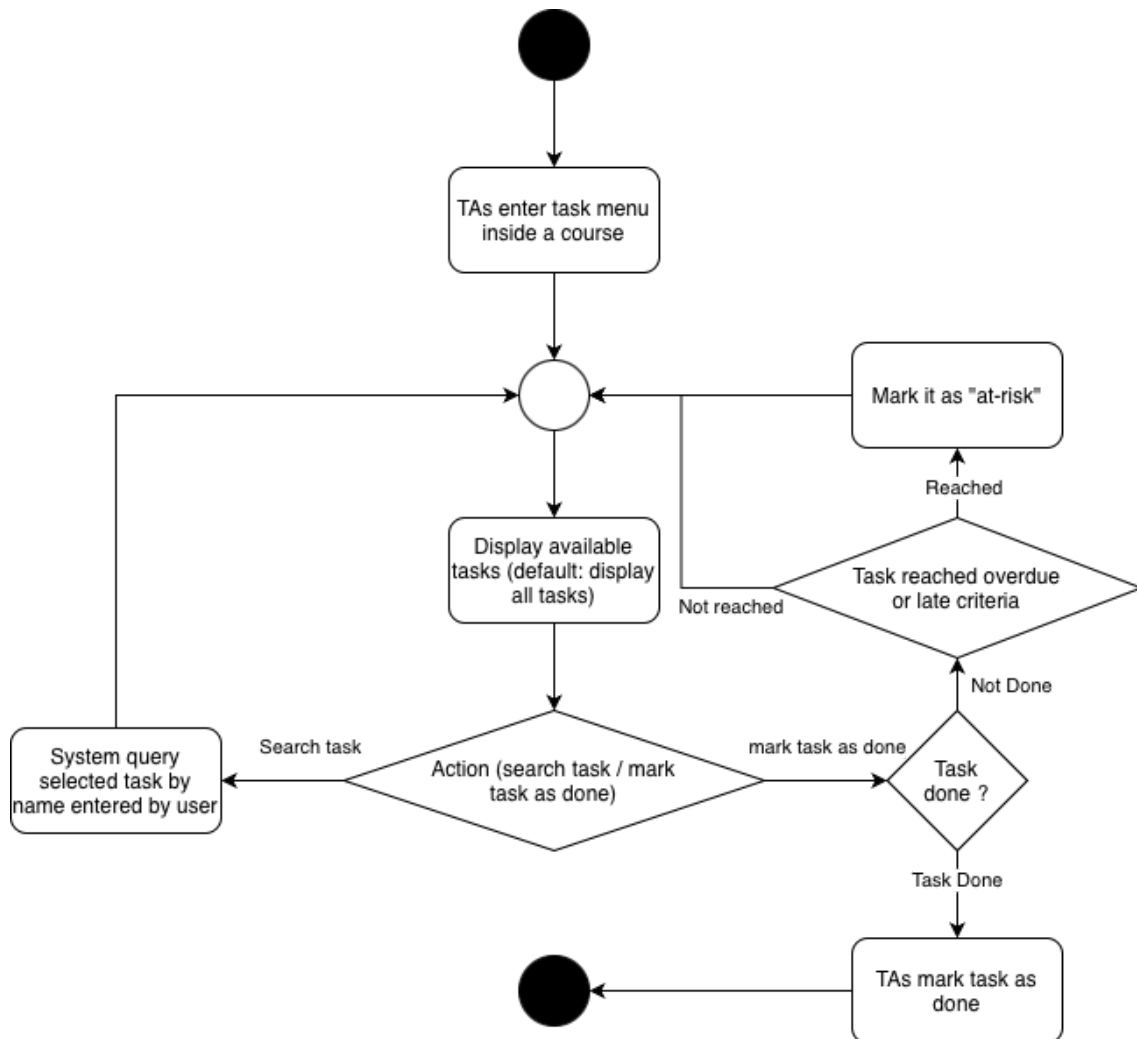
AD-4: Course's tasks creation

This diagram captures the process of course's task creation, used to derive SRS-4.



AD-5: TAs task handling

This diagram demonstrates the process where TAs handle incoming tasks that have been assigned by the lecturer, used to derive SRS-6, SRS-10, SRS-12.



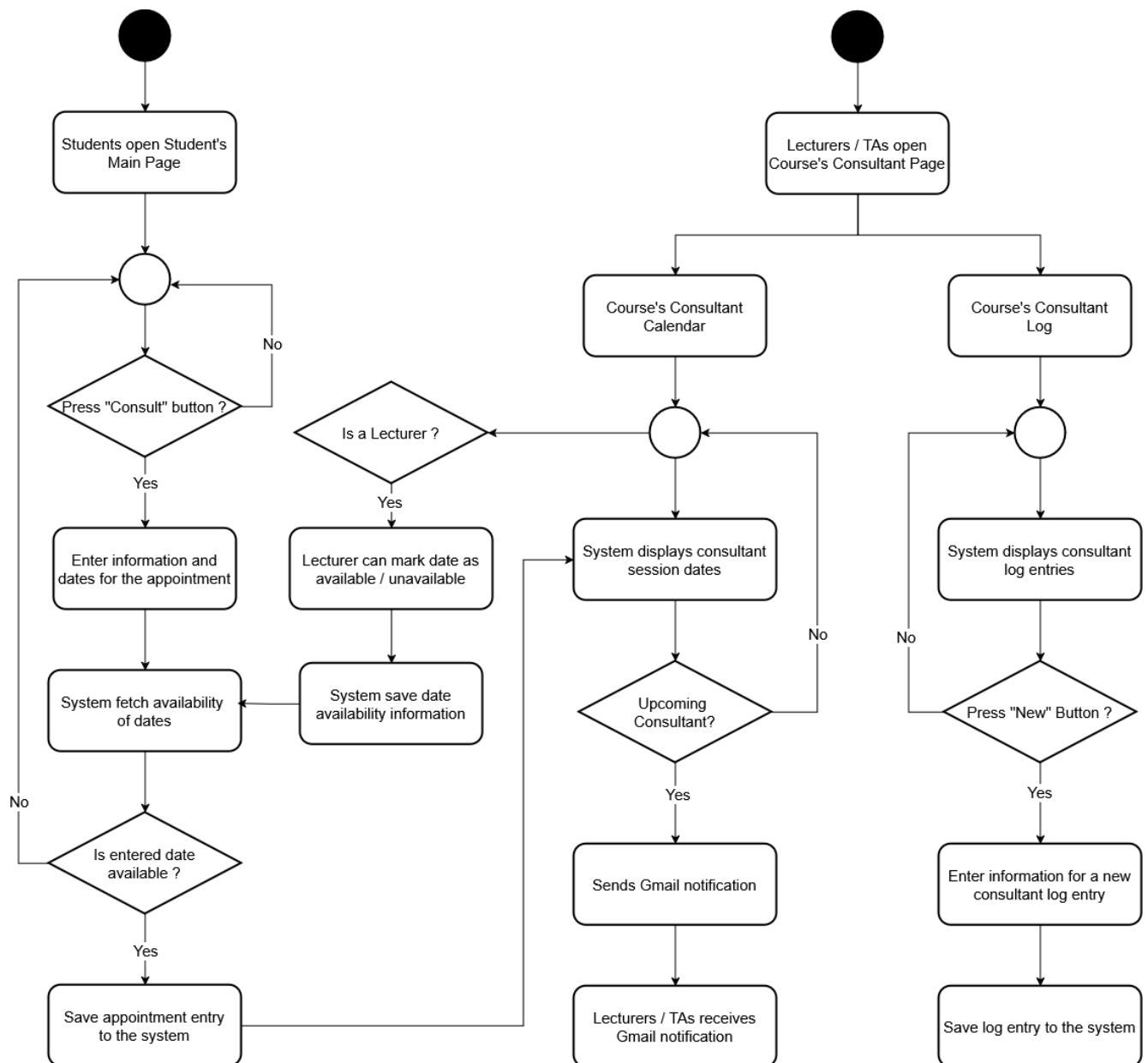
AD-6: Course's tasks management

This diagram captures the process of course's task management including update and delete function, used to derive SRS-5.



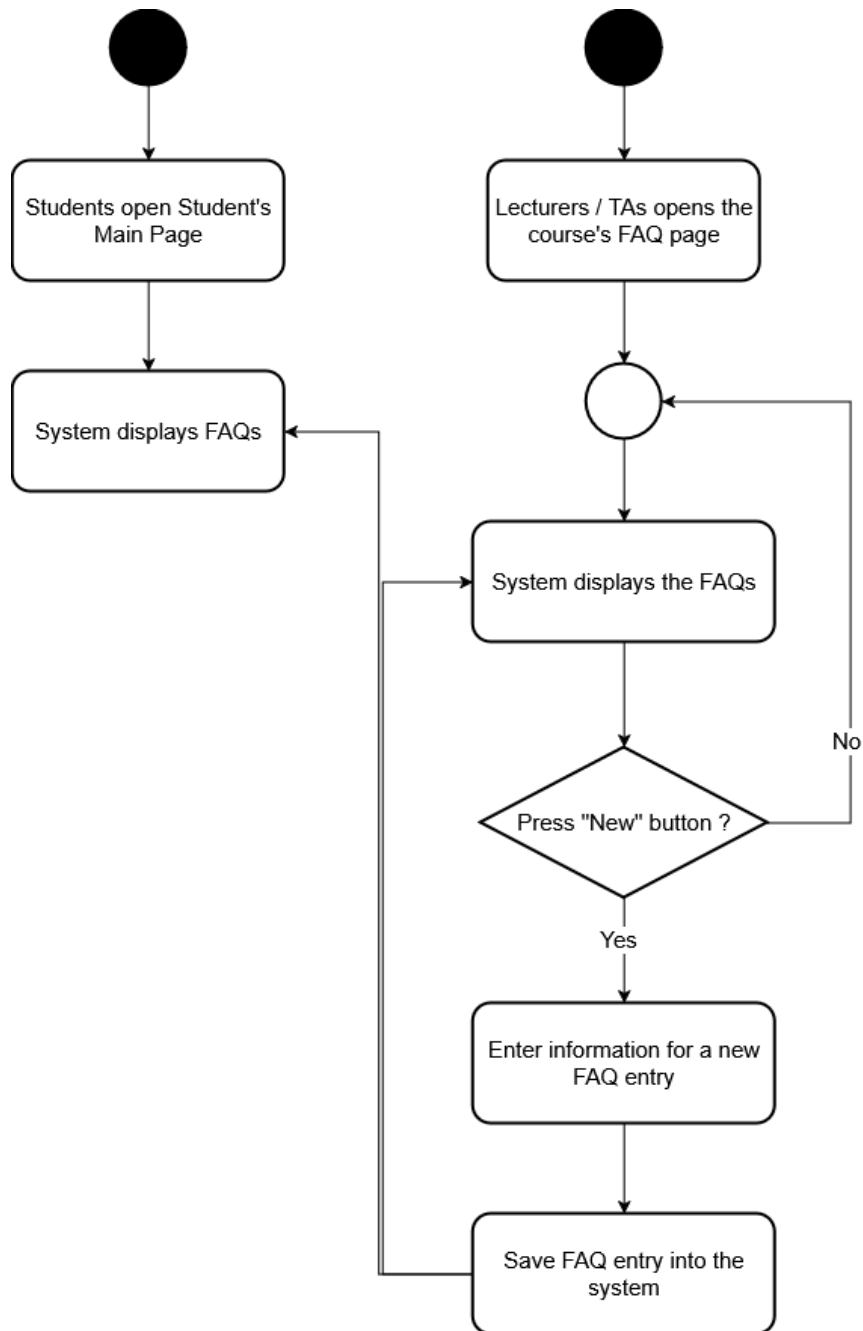
AD-7: Consultant Appointment and Logging

This diagram captures the confirmation and archive flow used to derive SRS-8 and SRS-9.



AD-8: FAQ Log

This diagram captures the confirmation and archive flow used to derive SRS-7 and SRS-9.



8. System Requirement Specification

ID	System Requirement
SRS-1	The system shall allow registered users to create and manage their own course instance.
SRS-2	The system shall let course instance owners add/manage other Lecturers and TA accounts.
SRS-3	The system shall allow the Lecturer to create an announcement to a course to notify TAs, and allow both Lecturers and TAs to post comments under course announcements.
SRS-4	The system lets Lecturers create and assign tasks with due dates to TAs to track the progress of TAs.
SRS-5	The system lets Lecturers manage available tasks that have been query into database (update content, delete)
SRS-6	The system shall let TAs show the progress of the assignment to allow the Lecturer to see the working progress.
SRS-7	The system shall only allow Lecturers and TAs to create FAQs log for individual courses. Students will be given access to only a static page.
SRS-8	The system shall allow students to create a consultation appointment and notify TAs with the appointment date.
SRS-9	The system shall allow TA to log the consultation for further review when creating FAQs.
SRS-10	The system shall aggregate all pending tasks and schedule and provide overview information for all registered users.
SRS-11	The system shall let any user with @ ku.th email holder to register and login into the system.
SRS-12	The system shall look out for any overdue task or untouched task and marked it as at-risk tasks.

9. Software Architecture

The recommended architecture for this system is a Modular Monolithic Architecture using the Model-View-Controller (MVC) pattern. The system will be developed as one application, but its features will be separated into different modules. This makes the system easier to develop, test, and maintain.

Rationale

The system is being built and maintained by a small development team and the current scope covers course management (SRS-1, SRS-2), announcements (SRS-3), task **tracking (SRS-4, SRS-5, SRS-6)**, FAQ management (SRS-7), consultations (SRS-8), dashboard aggregation (SRS-9). A traditional monolithic architecture would prevent the spaghettiness of the codebase and provide access control for each feature. Modular monolithic architecture would also allow each member to create their own module without stepping on others code. A microservices would present the complexity without any advantages for us since we aren't operating in the 1M+ requests. But if push comes to shove in the future, the modular monolithic architecture will make converting to microservices much easier than traditional monolithic.

Component	Layer/Type	Description
Web/Mobile Client	Presentation (View)	Displays the dashboard and course information to Lecturers, TAs,. It also shows announcements, tasks, FAQs, consultation information, appointments for students.
Courses and Controller	Controller	Receives requests related to users and courses, checks user permissions, and sends the requests to the appropriate business logic.

Announcement Controller	Controller	Handles creating and viewing course announcements and comments. It checks whether the user has permission to create or manage the course announcement.
Task Controller	Controller	Handles creating, assigning, updating, and deleting tasks. It also manages task status and due dates.
Consultation and FAQs	Controller	Handles consultation appointments, and FAQs creation and viewing, for students for students.
Dashboard Controller	Controller	Display Lecturer/TAs courses, schedules, tasks, for personalized overview.
Business Logic & Models	Model	Contains the rules for users, courses, announcements, tasks, consultations, FAQs, and dashboards. It validates data and controls how the system processes for each request.
Database	Persistence	Stores users, courses, announcements, comments, tasks, schedules, consultations, and FAQs title/comments. The system uses Supabase PostgreSQL as the database and SQLAlchemy ORM to interact with the data.

Authentication/Authorisation Module	Cross-cutting	Handles user login and checks for permission for Lecturer or TA. It ensures that each permission can only access features and data that they are allowed to use.
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10. Data Storage Technology

The ideal data storage technology is Supabase, which provides a PostgreSQL-based relational database as the single data store for this iteration. SQLAlchemy ORM is used as the application's database abstraction layer to define models, manage relationships, and interact with the Supabase PostgreSQL database.

Rationale

The data in the SRS is structured and has a fixed schema, including Users, Courses, Tasks, Announcements, Consultations, and FAQs. PostgreSQL supports relationships, data validation, and transaction consistency, making it suitable for this system. Supabase provides a managed PostgreSQL database, while SQLAlchemy ORM simplifies database access and model management. This keeps the technology stack simple and suitable for the current system.

Component	Data Storage Technology	Description
TALecturer Database	Supabase PostgreSQL with sqlalchemy	Includes Users, Courses, Tasks, FAQs Appointment, relational table storing related records and columns.

11. Development Environment

Development and deployment environments will be standardized using Docker containerization to ensure platform independence, operational consistency, and streamlined team collaboration.

Rationale

Using a containerized environment ensures consistency across all team members' local machines and deployment hosts, eliminating the "works on my machine" issue. By packaging application dependencies and runtime configurations into isolated containers, the project can be deployed smoothly on shared machines without requiring complex manual setup or causing software conflicts with existing system tools. Additionally, this setup decouples local execution from storage by allowing the containerized application to establish secure connection strings directly to the isolated self-hosted Supabase PostgreSQL database.

Component	Development Environment	Description
Application Server (Controller + Model)	Docker Container	Runs the frontend/backend on the docker container on the developer machine.
Users, Courses, Tasks, FAQs, Schedule, Appointment Tables for Database	Self hosted supabase via Docker	The database itself is hosted on the same machine as the application, but isolated via docker container.

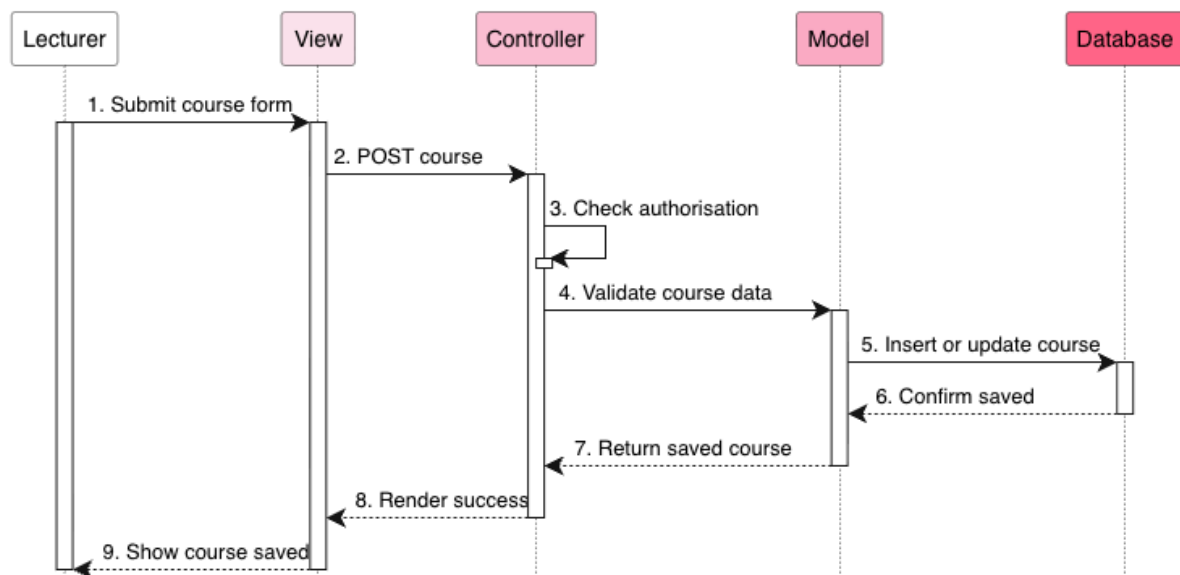
12. Sequence Diagrams for All SRS

The following sequence diagrams are derived from the chosen MVC modular architecture (Section 1) and cover all four use cases traced in the SRS traceability matrix:

- UC-1 Course creation/management,
- UC-2 Announcement creation/management,
- UC-3 UC-3 course task management (create/update/assign/delete),
- UC-4 FAQs creation,
- UC-5 Students consultation appointment with TAs/Lecturer, and UC-6 Lecturers and TAs personalized overview

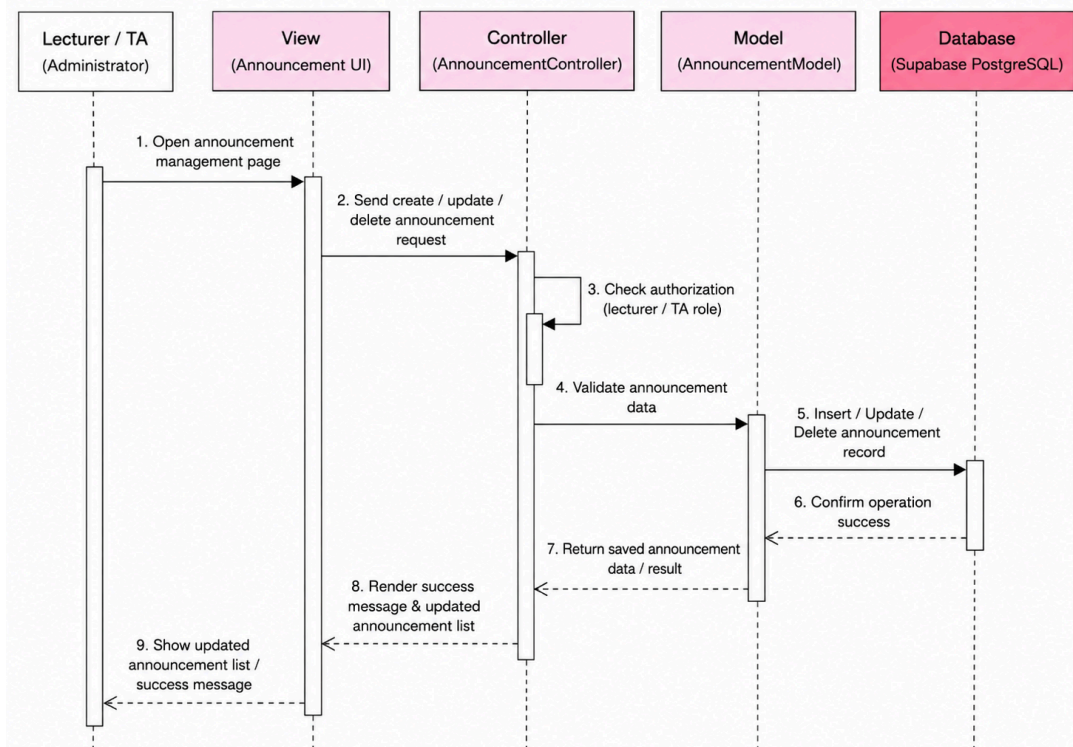
SQD-1: Course Creation and Management

Traced requirements: SRS-1, SRS-2



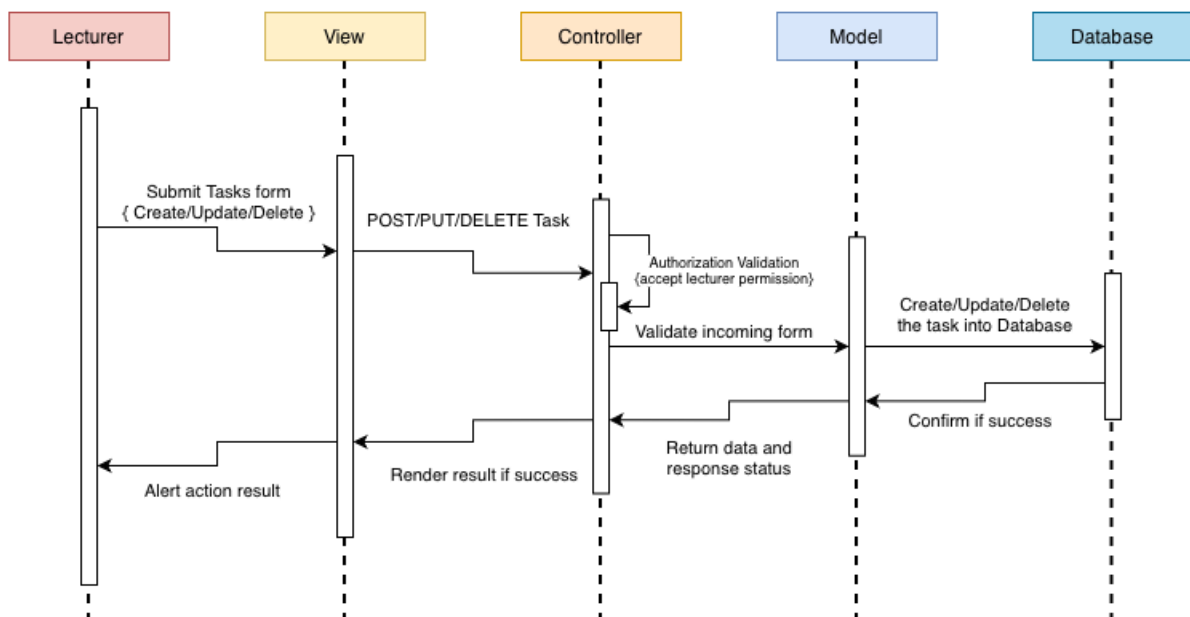
SQD-2: Announcement Creation and Management

Traced requirements: SRS-3



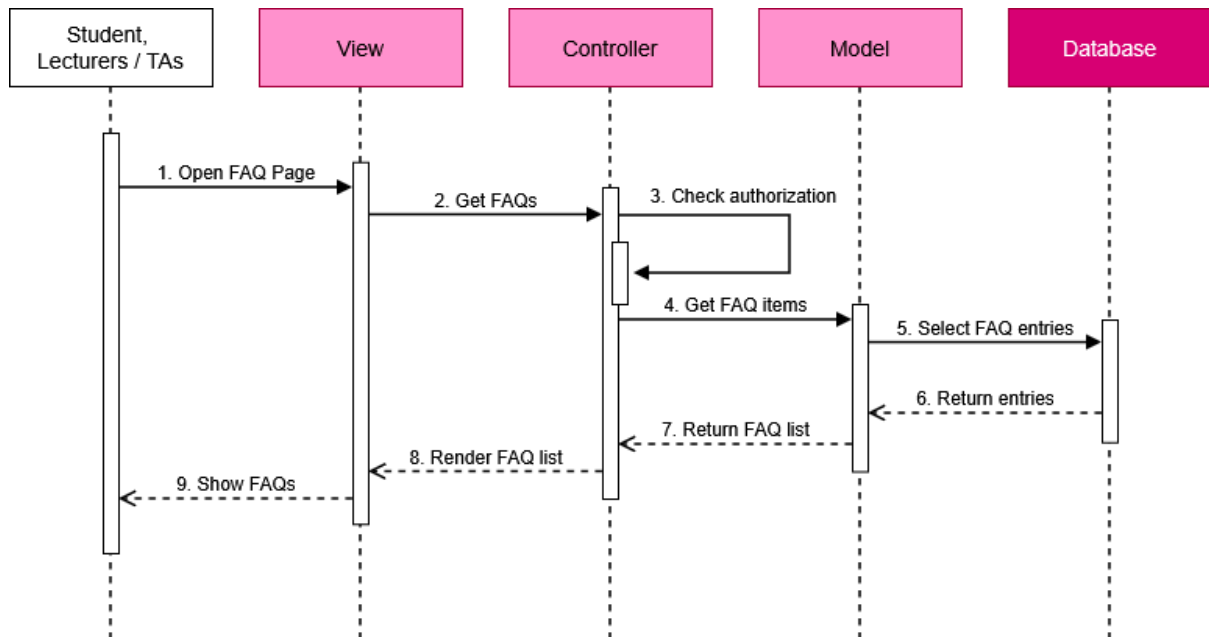
SQD-3: Course tasks management

Traced requirements: SRS-4, SRS-5, SRS-6, SRS-12



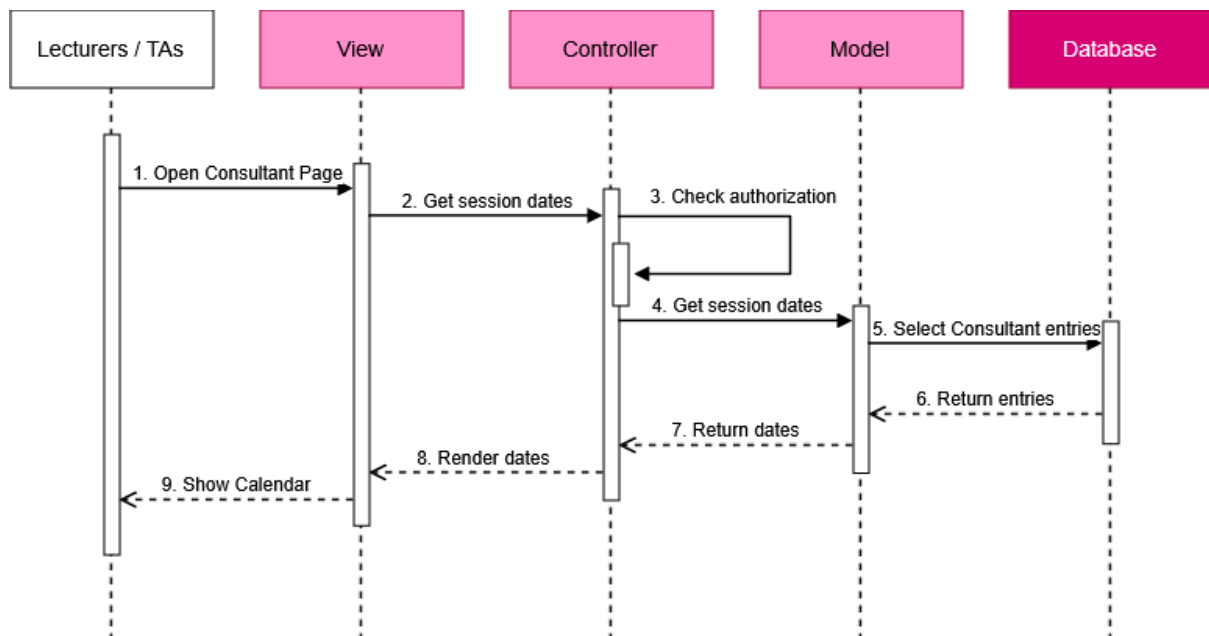
SQD-4: FAQs creation

Traced requirements: SRS-7, SRS-9



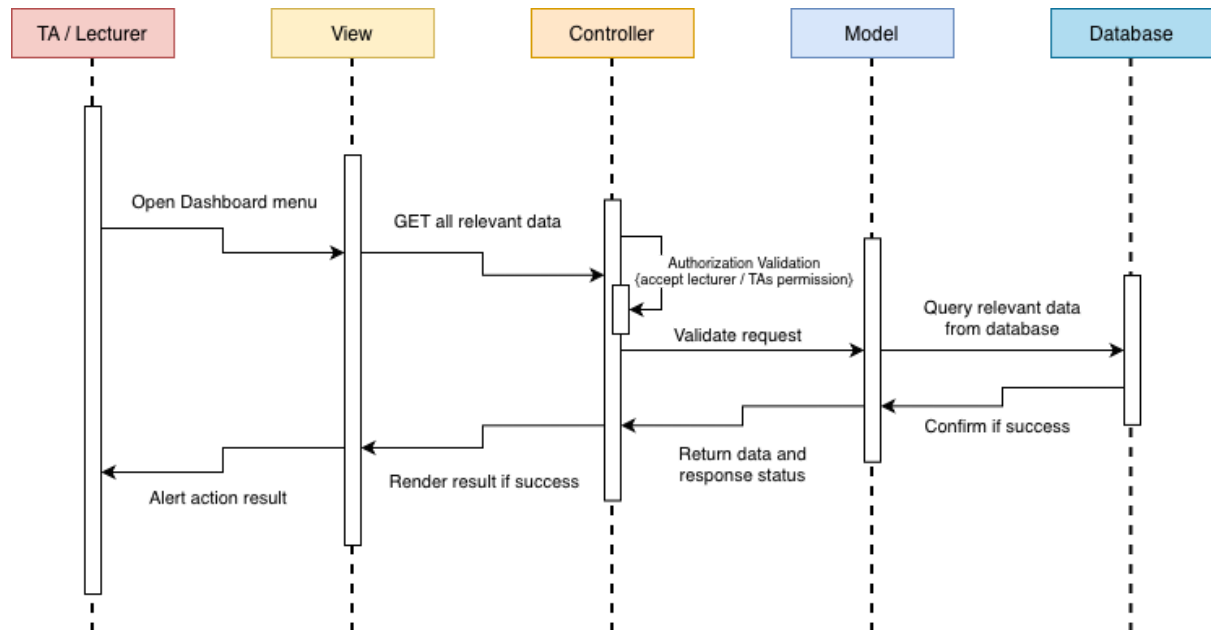
SQD-5: Consultation appointment and Personalized Calendar

Traced requirements: SRS-8



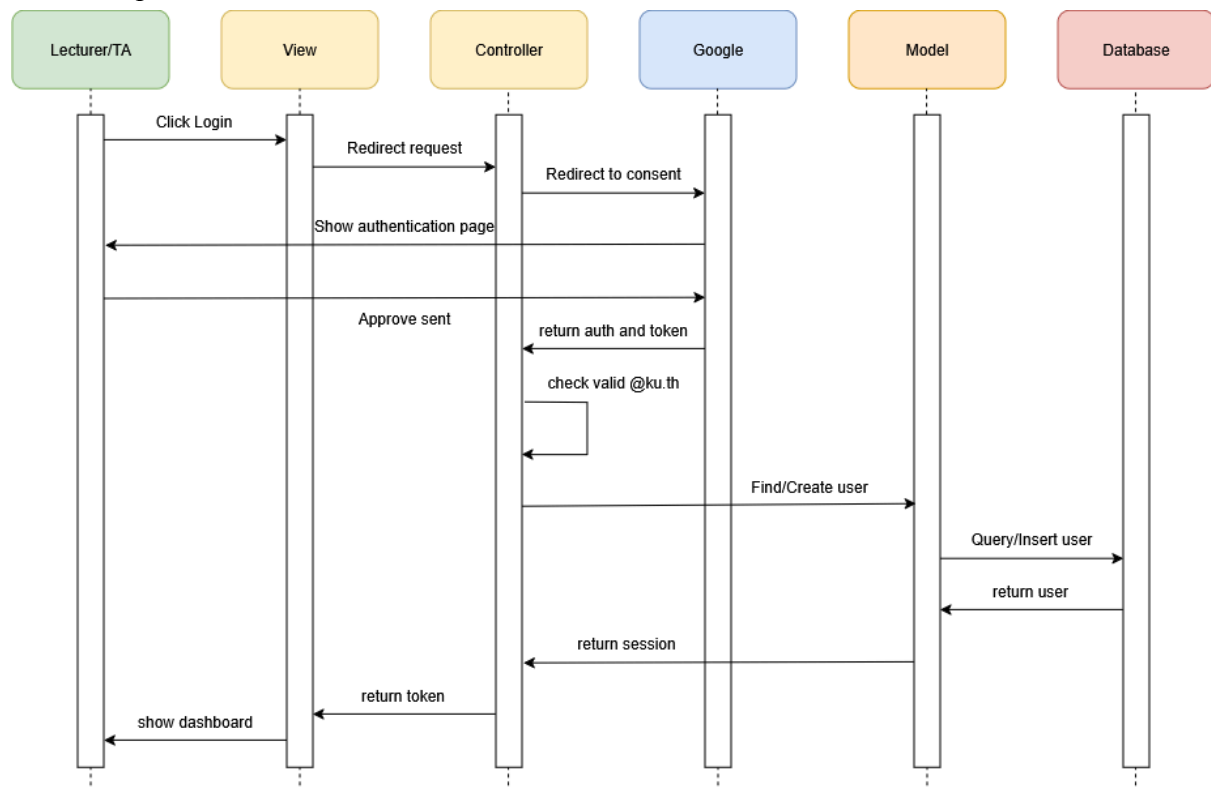
SQD-6: Lecturer/TAs Overview dashboard

Traced requirements: SRS-10



SQD-7: Google Authentication to Dashboard

Traced requirements: SRS-11



13. Traceability Matrix

Sub-Goal	Use Case	User Story	URS	Activity Diagram	SRS
SG-1	UC-1 Course creation /management	US-1	URS-1	AD-2	SRS-1, SRS-2
SG-2	UC-2 announcement creation/management	US-2	URS-2	AD-3	SRS-3
SG-3	UC-3 course task management (create/update/assign /delete)	US-3 US-4 US-10	URS-3	AD-4, AD-5, AD-6	SRS-4,SRS-5, SRS-6, SRS-12
SG-4	UC-4 FAQs creation	US-5	URS-5	AD-8	SRS-7, SRS-9
SG-5	UC-5 Consultation with appointment with TAs/Lecturer, and personalise calendar display	US-6 US-7	URS-4	AD-7	SRS-8
SG-6	UC-6 Lecturers and TAs personalized overview	US-8 US-9	URS-6	AD-2	SRS-10
SG-7	UC-7 Authentication via Google Account	US-11	URS-7	AD-1	SRS-11

From SRS to Sequence Diagram

SRS	SQD
SRS-1	SQD-1
SRS-2	SQD-1
SRS-3	SQD-2
SRS-4	SQD-3
SRS-5	SQD-3
SRS-6	SQD-3
SRS-7	SQD-4
SRS-8	SQD-5
SRS-9	SQD-4
SRS-10	SQD-6
SRS-11	SQD-7
SRS-12	SQD-3